Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/23/2009
PI Name:	Wang, Ya M.D., Ph.D.		
Project Title:	The mechanism of excess relative risk on carcinogenesis induced by high-LET radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
PI Email:	ywang94@emory.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(404) 778-1832
Organization Name:	Emory University		
PI Address 1:	School of Medicine Radiation Oncology		
PI Address 2:	1365 Clifton Road NE		
PI Web Page:			
City:	Atlanta	**State:**	GA
Zip Code:	30322	**Congressional District:**	5
Comments:			
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AF24G		
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Task Description:	Our hypothesis in this application is that high LET radiation induced excess relative risk on mutagenesis as well as on carcinogenesis is due to inefficient Ku-dependent non-homologous end-joining (NHEJ) repair. The results from this project will provide a mechanism to explain why high LET radiation induced such excess relative risk (ERR), and will significantly improve our understanding of the relationship between DNA repair and high LET radiation induced such ERR. Therefore, the results from this proposal will assist NASA Space Radiation Program to reach its major goal by providing specific parameters for evaluating high LET radiation induced ERR on mutagenesis as well as on carcinogenesis and by providing protective targets for reducing high LET radiation induced mutagenesis as well as carcinogenesis.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study described in this application will determine whether the major reason for high linear energy transfer (LET) ionizing radiation (IR) to induce excess relative risk (ERR) as compared with low LET IR on carcinogenesis is because of inefficient Ku-dependent non-homologous end-joining repair. The results from this proposal will answer the question whether inefficient Ku-dependent repair is the major reason for high LET IR to induce ERR on carcinogenesis. Results from this proposal will assist the NASA Space Radiation Program to reach its major goal by providing specific parameters for evaluating the ERR on carcinogenesis and by providing specific protective targets for reducing high LET IR-induced ERR on the carcinogenesis.
Task Progress:	Introduction: This project entitled "The mechanism of excess relative risk on carcinogenesis induced by high-LET radiation" (NNX07AT40G) was funded in September of 2007. Due to the PI relocating from Thomas Jefferson University to Emory University on September 1, 2008, the original grant (NNX07AT40G) was closed in Thomas Jefferson University and restarted (NNX09AF24G) in Emory University. This report covers the period (2008 September 1-2009 August 31): This project, in response to the NASA Program NNJ07ZSA001N, addresses research related to reducing the risks of high LET radiation-induced cancer. High linear energy transfer (LET) ionizing radiation (IR), when compared with low LET IR, induces more death in vitro as well as in vivo, more mutations in vitro and more cancers in vivo. IR-induced DNA double strand breaks (DSBs) are the severest threat for IR-induced killing as well as the initial and the essential event for IR-induced genomic integrity. Two kinds of major DSB repair exist in eukaryotic cells, homologous recombination repair (HRR) and non-homologous end-joining (NHEJ). HRR is the major pathway for repairing DSBs in yeast, but both the NHEJ and HRR pathways are required for DSB repair in mammalian cells. NHEJ repair includes a Ku-dependent main pathway and a PARP-1 dependent complementary pathway. High LET IR induces more cell killing when compared with low LET IR and is described as relative biologic effectiveness (RBE), which is believed to be caused by ineffective rejoining of DNA DSBs involved in the Ku-dependent NHEJ pathway. However, it remains unclear whether high LET IR also affects the pathways of HRR and PARP-1-dependent complementary NHEJ. Our recent data suggest that high LET IR affects only the Ku-dependent main NHEJ but not other repairs, including HRR and PARP-1-dependent complementary NHEJ. Based on our new data, we hypothesize that the major reason for high LET IR to induce excess relative risk on carcinogenesis is because of inefficient Ku-dependent non-homologous end-joining (NHEJ) repair. The purpose of this project is to test our hypothesis and three specific aims are proposed to accomplish this goal. A.1. Determine whether the major reason for high LET IR inducing excess deaths in both cells and mice is because of inefficient Ku-dependent NHEJ repair. A.2. Determine whether the major reason for high LET IR inducing excess relative risk (ERR) on carcinogenesis in mice is because of inefficient Ku-dependent NHEJ repair. A.3. Determine whether the major reason for high LET IR inducing excess relative risk (ERR) on genomic instability is because of inefficient Ku-dependent NHEJ repair. We will use the repair deficient mice combining in vitro and in vivo approaches to carry out our experiments. Regardless of whether our hypothesis is proven to be right or wrong, results from these experiments will tell us how much a percentage of each repair pathway (Ku-dependent NHEJ, PARP-1-dependent NHEJ or HRR) is responsible for the high LET IR induced ERR on genomic instability, and will provide valuable information for predicting high LET IR induced carcinogenesis. Progress and Impact: This project includes two major categories: Animal experiment and cells experiment. We report the progress from these two categories. 1. Animal work: Although the project progress was affected due to the PI's laboratory relocation, (for example, we could not make the BNL 2008 fall run because of waiting for animal protocols to be approved) we still managed to initiate irradiating the cells and mice in BNL in spring 2009. The animal work requires several BNL beam times, which is planned to be finished in 2010. 2. Cells work: We carried out some experiments by using different types of cells and obtained some interesting results: 1). S-phase cells are more sensitive to heavy ion, the high linear energy transfer (LET) radiation (component of space radiation), although they are more resistant to x or gamma ray. After further study of the mechanism, we believe that the phenotype of S-phase sensitive to high-LET radiation is due to the fact: high-LET radiation could generate much more small fragments (< 40 base pairs) that result in inefficient Ku binding in S –phase, which further supports our hypothesis: high-LET killing more cells when compared with low-LET radiation is because of inhibition of Ku-dependent NHEJ. 2). We previously reported that DNA double strand break (DSB) activated checkpoint mainly facilitates homologous recombination repair (HRR) and has little effect on non-homologous end-joining (NHEJ). Combining our new finding with high-LET radiation we tested our hypothesis that checkpoint response plays a more protective role in HZE particle-irradiated cells than in X ray-irradiated cells and obtained supportive results. 3). It has been known for more than two decades that low dose radiation (< 0.1 Gy) could induce adaptive responses. The adaptive responses could protect cells and animals from high challenge dose induced killing and carcinogenesis. Because DNA DSBs induced by radiation are the severest threat for cell survival and genetic integrity, the low dose adaptive responses are believed to promote DNA repair. However, it remains unclear which DNA DSB repair pathway: NHEJ or HRR is promoted by the low dose radiation. We found that low dose radiation could not induce any protective role in high-LET irradiated cells. After combing these results with our repair theory about high-LET radiated cells, we hypothesized that low dose radiation induced adaptive responses are through promoting NHEJ. We tested this hypothesis and obtained supportive results. These results further support our hypothesis that the major reason for high LET IR to induce excess relative risk on carcinogenesis is because of inefficient Ku-dependent non-homologous end-joining (NHEJ). Because our results are provide mechanism information to elucidate the nature different between high-LET radiation and low-LET radiation, these results will contribute to both estimating and reducing the risk of space radiation to the astronauts' health.
Bibliography Type:	Description: (Last Updated: 07/07/2021)

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